DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

2. TITLE	1. IDENTIFICATION NUMBER
Laminating Process Description (Boats)	DI-MISC-80703

3. DESCRIPTION / PURPOSE

3.1 The Laminating Process Description delineates the procedure used in fabricating fiber reinforced plastic boats in accordance with the drawings and specifications applicable to the contract. This to insure that the contractor's process meets the specification requirements.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
881115	N/PMS300		

7. APPLICATION / INTERRELATIONSHIP

7.1 This data item description (DID) contains format and content preparation instructions for the data product generated by the specific and discrete task requirements as delineated in the contract.

7.2 This DID applicable to the construction of fiber reinforced plastic hulls.

7.3 This supersedes UDI-E-23086.

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		N4568

10. PREPARATION INSTRUCTIONS

10.1 LAMINATING PROCESS PROCEDURE DESCRIPTION

10.2 General

a. For each of the items below, describe the quality control and inspection procedures by which compliance with the requirements of the specification will be assured, or refer to the appropriate section of the contractor's quality assurance or inspection document. Include job titles, inspection intervals, and record keeping procedures.

b. Where information concerning methods is required, include tools, equipment, procedures, and quality controls used.

10.2.1 For each of the materials used in lamination and assembly, identify:

a. Manufacturer

b. Specification (by part or model number, or by Federal, Military, or commercial specification)

c. Method of storage (protection from heat and weather, temperature and humidity control)

11. DISTRIBUTION STATEMENT

Distribution Statement A: Approved for public release; distribution is unlimited.

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10. PREPARATION INSTRUCTIONS (Continued)

10.2.2 Laminating Process:

a. Describe the methods of temperature and humidity control and monitoring in the laminating facility.

b. Identify the methods of preparation of the molds for lamination. Include repair polishing methods, type, and method of application of mold release agents.

c. Provide a chronology of the hull laminating process from the application of the gel coat to the beginning of the application of the hull internal structure. Include the time elapsed from the application of the gel coat, and the time and number of persons required to complete each operation. Include time and duration of breaks in the process, and identify use of scaffolding and rolling of the mold.

d. Identify the method of application of the gel coat, the gel time used, and the method by which thickness is monitored and controlled.

e. Identify when and by what methods the promoter, accelerator, catalyst, thixotrope and other gel coat and laminating resin additives, as applicable, will be measured and mixed into the resin.

f. Identify the gel times used with the laminating resin at each stage of the laminating process, and the means by which the gel time will be monitored and controlled.

g. Describe the method of application of the reinforcing material and laminating resin. Include direction of lay-up, location of laps, precutting and templating of operations used, methods used to wet out and consolidate the laminate and to remove air, and methods used to monitor and control resin content. Provide sketches detailing treatment of laps and doubler at chines, keel, transom corners,stem, and doublers.

h. If a cored laminate is used, describe the type of core and whether solid, scored, perforated or otherwise machined to improve conformity to the laminate. Describe the method used to precoat the core. Describe the method of preparing and bedding the core in the laminate and of assuring complete contact of the core with the laminate. If a syntactic foam bedding material is used, describe the methods used to mix, catalyse and apply the bedding foam, and the means by which the gel time will be monitored and controlled. Describe the methods by which complete contact of the core with the bedding foam will be monitored and maintained.

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10. PREPARATION INSTRUCTIONS (Continued)

i. Describe methods used to install secondarily bonded structures and doublers. Describe surface preparation and cleaning methods.

j. Identify the surface sealing resin used. Describe method of application, means of achieving surface cure, and means of controlling thickness.

10.2.3 Assembly:

a. Describe the part at time of removal from the mold. Identify the structural components such as bulkheads and framing which are installed, Barcol readings taken, and whether surface sealing resin has been applied.

b. Describe the methods used to remove the part from the mold.

c. Describe laminate inspection and repair procedures.

d. Describe edge trimming and sealing procedures.

e. Describe the method of preparation for final assembly and final assembly of major structural connections. Include the hull-deck connection and other major structural connections, if used.

f. Describe the method of storage and protection of parts prior to final assembly, finish painting and delivery.

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